

PDF file name: IEEE-2020-Adaptive Fuzzy Finite-Time Tracking

Original article word count: 2939

Extracted content word count: 1304 (44% of the original word count)

Title of the article: *Adaptive Fuzzy Finite-Time Tracking Control of Uncertain Non-Affine Multi-Agent Systems With Input Quantization*

The content extracted is as follows:

Over the past twenty years, the cooperative control problem of multi-agent systems has paid extensively attention in many fields, such as multiple unmanned aerial vehicles, flexible robot manipulator systems, wireless sensor networks and large-scale systems [1] [6].

In the existing literature, there are many results on the first- and second-order multi-agent systems with undirected graph, directed graph and switching graph have been obtained, see [7] [12] and references therein. For the most practical systems, however, they need to be modeled as the high-order system rather than simple first- or second-order system.

Moreover, the relevant results on first- and second-order multi-agent systems cannot be used in high-order multi-agent systems.

The control problem of high-order multi-agent systems has been considered in some literature.

In [13], a reduced-order observer was presented to solve the distributed tracking control problem of high-order nonlinear multi-agent systems with heterogeneous leader.

In [14], a distributed region regulator was presented to deal with the synchronization problem of a class of high-order multi-agent systems.

For the high-order nonlinear strict-feedback multi-agent systems, the authors of [15] used the distributed extended state observer and backstepping technique to solve the practical time-varying output formation tracking problem, and in [16], [17], the consensus tracking problem of high-order multi-agent systems with high-dimensional leader was firstly considered, where a distributed robust adaptive consensus control protocol with local observer was designed. On the basis of the results of [13] [17], the containment tracking problem of multi-agent systems with pure-feedback form was analyzed in [18], where the distributed adaptive fuzzy control protocol and local quantized controller were considered. Based on the advantages of the fuzzy control method, some control problems are well solved by designing the control law based on with the fuzzy control method [19], [20].

To the best of author's knowledge, however, only a few results focus on the control problem of nonaffine multi-agent systems.

For instance, in [21], the agreement control of a class of nonaffine multi-agent systems was addressed, where a distributed neural adaptive consensus control scheme with guaranteed performance was designed. In [22], [23], the tracking problem of uncertain nonlinear nonaffine multi-agent systems was studied, and the distributed output-feedback control protocol with prescribed tracking performance and the distributed containment controller were investigated, respectively.

Referring to the literature of above-mentioned, it is easy to find that the control problem of multi-agent systems is based on the case that the control time is close to infinite.

In some cases, it is desirable that the control objective of a multi-agent system be achieved in a finite time.

Recent years, the finite-time control problem of multi-agent systems has been greatly developed. The finite-time control scheme for the attitude tracking problem of the spacecraft systems was investigated in [24].

Considering the input saturation, a sliding mode observer was designed to solve the global finite-time tracking problem of second-order multi-agent systems in [25].

For the heterogeneous nonlinear high-order multi-agent systems with mismatched disturbances, the authors of [26] designed the families of consensus protocols and solved the finite-time output consensus problem.

Furthermore, the containment control of a class of nonaffine pure-feedback multi-agent was addressed in [27], where a distributed neural adaptive control strategy was proposed. Moreover, the finite-time consensus tracking control of a class of nonlinear multi-agent systems with prescribed performance was studied in [28], where the error constrained

control was applied to obtain satisfactory control performances.

In addition, the finite-time control problems of Markov jump systems and nontriangular stochastic nonlinear systems are analyzed in [29] [33].

Therefore, it is necessary to design a suitable control protocol for uncertain nonlinear multi-agent systems with input quantization to achieve the desired control objective in a finite time.

This paper investigates the finite-time tracking control problem of a type of nonlinear multi-agent systems with input quantization and uncertain nonlinearity.

A class of uncertain nonlinear multi-agent systems with input quantization is considered in this paper.

With the help of the differential mean value theorem, the nonlinear functions of multi-agent systems can be decomposed into a strict-feedback form.

Based on the fuzzy logic systems approximator and backstepping technique, an adaptive fuzzy finite-time tracking control protocol with adaptive update laws is designed. By the simulation analysis, the control protocol proposed in this paper can guarantee the tracking error of each follower agent converges to a small interval within a finite time.

In Section II, the preliminaries and problem formulation are introduced. The detailed process of adaptive fuzzy finite-time tracking control protocol design is given in Section III, and Section IV provides the simulation example to verify the effectiveness of the theoretical analysis.

Consider a class of uncertain nonlinear multi-agent systems with $N(N > 1)$ follower agents and one leader agent.

$$\dot{v}(t) = cv^{\mu}(t) + d, \quad t \geq 0$$

According to the multi-agent systems (3), the tracking error of follower agent i can be designed as $e_i = y_i - y_0$ ($i = 1, \dots, N$) and y_0 is the output of leader agent, then one has $\text{Let } z_{i,1}$ be the local consensus error of follower agent i , hence, we have: where $1 = [1, \dots, 1]$.

It implies that the tracking error of each follower agent converges to a small neighborhood of the origin and remains there after the finite time t .

The proof is completed. Remark 1: Compared with the traditional methods which the convergence time is usually t , we can obtain convergence in a finite time, which implies that the control method designed in this paper has a better convergence rate.

Consider a class of uncertain nonlinear multi-agent systems with four follower agents and one leader agent, where each follower agent represents an inverted pendulum system and the leader is used as a signal generator to generate tracking signal.

The output response curves with four follower agents and one leader agent under the general in finite-time control scheme and the finite-time control scheme are shown as Figure 2.

The reason is that the following four agents can track the leading agent in a finite time, which shows that the fuzzy logic system and adaptive parameters also converge in a finite time.

Although the control input is quantized, the tracking control problem with the finite-time control law designed in this paper can be achieved. In addition, the response curves for the adaptive update laws and are displayed as Figures 5-6.

In order to verify the effectiveness of the designed control law (44) (scheme 1), a comparative study with the finite-time control law (scheme 2) designed in literature [28] is given in this paper.

According to the Figure 7, the control scheme 1 designed in this paper and the control scheme 2 designed in [28] can realize the tracking control problem of given multi-agent system in finite time.

Compared with the scheme 2, however, the control law designed in this paper can achieve the control result in a shorter time, and the control is better than scheme 2.

This paper has investigated the finite time tracking control problem of a type of uncertain nonlinear multi-agent systems.

Compared with the existing multi-agent systems, a more general model with unknown nonlinear functions and uncertain nonlinearity has been considered. The control input of multi-agent systems has been quantized by the quantizer.

With the help of a fuzzy logic systems and the backstepping technique, furthermore, the fuzzy finite-time tracking control protocol has been proposed. Based on the presented control protocol, it has been proved that the tracking error of all follower agents can be held on a small enough neighborhood in a finite time.